

M003 Week 5 — English Worksheet (Beginner)

Topic: Into 3D (x, y, z) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

You know the wall (x, y) and the floor (x, z). This week you use **all three numbers at once: (x, y, z)**. x is across, **y is up (height)**, z is forward. Now you can put a block **anywhere** in space.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
set y to 0
repeat 5 times:
  place stone block at (3, y, 3)
  add 1 to y
```

The middle number y grows by 1 each time. Does this build a tower going up? Circle one: yes · no

Why?


```
place gold block at (2, 6, 2)
```

y is 6 and there is nothing below it. Is the gold block high in the air? Circle one: yes · no

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The tower should go **up**, so the **middle** number (y) must grow.

```
# clean
set y to 0
repeat 4 times:
  place stone block at (3, y, 3)
  add 1 to y
```

```
# buggy
set z to 0
repeat 4 times:
  place stone block at (3, 0, z)
  add 1 to z
```

What is wrong? Does the buggy one go up, or forward?

Pair B — The tower should start **on the ground**, where y is 0.

```
# clean
set y to 0
repeat 4 times:
  place stone block at (2, y, 2)
  add 1 to y
```

```
# buggy
set y to 3
repeat 4 times:
  place stone block at (2, y, 2)
  add 1 to y
```

What is wrong? Where does the buggy tower start?

3 · Fill the Gap

The tower should grow **up into the sky**. The middle number must grow. One word is missing. Fill it in using the word bank.

```
set y to 0
repeat 5 times:
  place stone block at (3, y, 3)
  add 1 to ____
```

Word bank: y · x · z

Write the missing word:

4 · Tell Me What You Built 📷

Now switch to your homework world. Build a **tower** that goes up using a loop, then put a gold block floating in the air above it. When you finish, come back here.

Send a photo or video of your tower, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My tower is ... blocks tall.

The number that grew was **y**, which means ...

My floating block is at (... , ... , ...).

5 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while you show your tower in the world. Talk like you are teaching a friend. Try to use these words: **x, y, z, height, up.**

1. Show your tower from the bottom to the top.
2. Read your loop out loud and say which number grows.
3. Point at the floating block and say all three of its numbers.

Write what you will say in your video. You can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.